

Clojure Cheat Sheet (Clojure 1.9 - 1.12, sheet v56)

Documentation	
clojure.repl/	doc find-doc apropos dir source pst javadoc (foo.bar/ is namespace for later syms)
Primitives	
Numbers	
Literals	Long: 7, hex 0xff, oct 017, base 2 2r1011, base 36 36rCRAZY BigInt: 7N Ratio: -22/7 Double: 2.78 -1.2e-5 BigDecimal: 4.2M
Arithmetic	+ - * / quot rem mod inc dec max min +' -' *' inc' dec' (1.11) abs (clojure.math/) floor-div floor-mod ceil floor rint round pow sqrt cbrt E exp expm1 log log10 loglp PI sin cos tan asin acos atan2 == < > <= >= compare
Compare	bit-and bit-or bit-xor bit-not bit-flip bit-set
Bitwise	bit-shift-right bit-shift-left bit-and-not bit-clear bit-test unsigned-bit-shift-right (see BigInteger for integers larger than Long)
Cast	byte short int long float double bigdec bigint num rationalize biginteger
Test	zero? pos? neg? even? odd? number? rational? integer? ratio? decimal? float? double? int? nat-int? neg-int? pos-int? (1.11) NaN? infinite?
Random	rand rand-int (1.11) (clojure.math/) random
BigDecimal	with-precision
Unchecked	*unchecked-math* unchecked-add unchecked-dec unchecked-inc unchecked-multiply unchecked-negative unchecked-subtract
Strings	
Create	str format "a string" "escapes \b\f\n\t\r\" octal \377 hex \ucafe" See also section IO/to string
Use	count get subs compare (clojure.string/) join escape split split-lines replace replace-first reverse index-of last-index-of (1.11) (clojure.core/) parse-boolean parse-double parse-long parse-uuid
Regex	#"pattern" re-find re-seq re-matches re-pattern re-matcher re-groups (clojure.string/) replace replace-first re-quote-replacement Note: \ in #" is not escape char. (re-pattern "\s*\d+") can be written #"\s*\d+"
Letters	(clojure.string/) capitalize lower-case upper-case
Trim	(clojure.string/) trim trim-newline triml trimr
Test	string? (clojure.string/) blank? starts-with? ends-with? includes?
Other	
Characters	char char? char-name-string char-escape-string literals: \a \newline (more at link)
Keywords	keyword keyword? find-keyword literals: :kw :my.name.space/kw ::in-cur-namespace ::namespace-alias/kw
Symbols	symbol symbol? gensym literals: my-sym my.ns/foo
Misc	literals: true false nil

Collections	
Collections	
Generic ops	count bounded-count empty not-empty into conj (clojure.walk/) walk prewalk prewalk-demo prewalk-replace postwalk postwalk-demo postwalk-replace
Content tests	distinct? empty? every? not-every? some not-any?
Capabilities	sequential? associative? sorted? counted? reversible?
Type tests	coll? list? vector? set? map? seq? record? map-entry?
Lists (conj, pop, & peek at beginning)	
Create	() list list*
Examine	first nth peek .indexOf .lastIndexOf
'Change'	cons conj rest pop

Vectors (conj, pop, & peek at end)	
Create	[] vector vec vector-of mapv filterv
Examine	(my-vec idx) → (nth my-vec idx) get peek .indexOf .lastIndexOf
'Change'	assoc assoc-in pop subvec replace conj rseq update update-in
Ops	reduce-kv
Sets	
Create unsorted	#{} set hash-set
Create sorted	sorted-set sorted-set-by (clojure.data.avl/) sorted-set sorted-set-by (flatland.ordered.set/) ordered-set (clojure.data.int-map/) int-set dense-int-set
Examine	(my-set item) → (get my-set item) contains?
'Change'	conj disj
Set ops	(clojure.set/) union difference intersection select See also section Relations
Test	(clojure.set/) subset? superset?
Sorted sets	rseq subseq rsubseq

Maps	
Create unsorted	{ } hash-map array-map zipmap bean frequencies group-by (clojure.set/) index
Create sorted	sorted-map sorted-map-by (clojure.data.avl/) sorted-map sorted-map-by (flatland.ordered.map/) ordered-map (clojure.data.priority-map/) priority-map (flatland.useful.map/) ordering-map (clojure.data.int-map/) int-map
Examine	(my-map k) → (get my-map k) also (:key my-map) → (get my-map :key) get-in contains? find keys vals
'Change'	assoc assoc-in dissoc merge merge-with select-keys update update-in (clojure.set/) rename-keys map-invert (1.11) (clojure.core/) update-keys update-vals GitHub: Medley
Ops	reduce-kv
Entry	key val
Sorted maps	rseq subseq rsubseq

Queues (conj at end, peek & pop from beginning)	
Create	clojure.lang.PersistentQueue/EMPTY (no literal syntax or constructor fn)
Examine	peek
'Change'	conj pop

Relations (set of maps, each with same keys, aka rels)	
Rel algebra	(clojure.set/) join select project union difference intersection index rename
Transients (clojure.org/reference/transients)	
Create	transient persistent!
Change	conj! pop! assoc! dissoc! disj! Note: always use return value for later changes, never original!
Misc	
Compare	= identical? not= not compare clojure.data/diff
Test	true? false? instance? nil? some?

Sequences	
Creating a Lazy Seq	
From collection	seq vals keys rseq subseq rsubseq sequence
From producer fn	lazy-seq repeatedly iterate (1.11) iteration
From constant	repeat range
From other	file-seq line-seq resultset-seq re-seq tree-seq xml-seq iterator-seq enumeration-seq
From seq	keep keep-indexed

Seq in, Seq out	
Get shorter	distinct filter remove take-nth for dedupe random-sample
Get longer	cons conj concat lazy-cat mapcat cycle interleave interpose
Tail-items	rest nthrest next fnext nnext drop drop-while take-last for
Head-items	take take-while butlast drop-last for
'Change'	conj concat distinct flatten group-by partition partition-all partition-by split-at split-with filter remove replace shuffle (1.12) partitionv partitionv-all splitv-at
Rearrange	reverse sort sort-by compare
Process items	map pmap map-indexed mapcat for replace seque

Using a Seq	
Extract item	first second last rest next ffirst nfirst fnext mnext nth nthnext rand-nth when-first max-key min-key
Construct coll	zipmap into reduce reductions set vec into-array to-array-2d mapv filterv
Pass to fn	apply
Search	some filter
Force evaluation	doseq dorun doall run!
Check for forced	realized?

Transducers (clojure.org/reference/transducers)	
Off the shelf	map mapcat filter remove take take-while take-nth drop drop-while replace partition-by partition-all keep keep-indexed map-indexed distinct interpose cat dedupe random-sample halt-when (1.12) partitionv-all
Create your own	completing ensure-reduced unreduced See also section Concurrency/Volatiles
Use	into sequence transduce eduction
Early termination	reduced reduced? deref

Spec (rationale, guide)	
Operations	valid? conform uniform explain explain-data explain-str explain-out form describe assert check-asserts check-asserts?
Generator ops	gen exercise exercise-fn
Defn. & registry	def fdef registry get-spec spec? spec with-gen
Logical	and or
Collection	coll-of map-of every every-kv keys merge
Regex	cat alt * + ? & keys*
Range	int-in inst-in double-in int-in-range? inst-in-range?
Other	nilable multi-spec fspec conformer
Custom explain	explain-printer *explain-out*

Predicates with test.check generators	
Numbers	number? rational? integer? ratio? decimal? float? zero? double? int? nat-int? neg-int? pos-int?
Symbols, keywords	keyword? symbol? ident? qualified-ident? qualified-keyword? qualified-symbol? simple-ident? simple-keyword? simple-symbol?
Other	string? true? false? nil? some? boolean? bytes? inst? uri?
scalars	uuid?
Collections	list? map? set? vector? associative? coll? sequential? seq? empty? indexed? seqable?
Other	any?

IO	
to/from	spit slurp (to writer/from reader, Socket, string with file name, URI, etc.)
...	
to *out*	pr prn print printf println newline (clojure.pprint/) print-table (clojure.pprint/) pprint cl-format also: (binding [*out* writer] ...)
to writer	format with-out-str pr-str prn-str print-str println-str
from *in*	read-line (clojure.edn/) read (clojure.tools.reader.edn/) read from reader
from reader	line-seq (clojure.edn/) read (clojure.tools.reader.edn/) read also: (binding [*in* reader] ...) java.io.Reader
from string	with-in-str (clojure.edn/) read-string (clojure.tools.reader.edn/) read-string
Open	with-open (clojure.java.io/) text: reader writer binary: input-stream output-stream
Binary	(.write ostream byte-arr) (.read istream byte-arr)
Misc	java.io.OutputStream java.io.InputStream GitHub: gloss byte-spec flush (.close s) file-seq *in* *out* *err* (clojure.java.io/) file copy delete-file resource as-file as-url as-relative-path GitHub: fs
Data readers	*data-readers* default-data-readers *default-data-reader-fn*
tap	(1.10) tap> add-tap remove-tap

Functions	
Create	fn defn defn- definline identity constantly memfn comp complement partial juxt memoize fnil every-pred some-fn
Call	apply -> ->> trampoline as-> cond-> cond->> some-> some->>
Test	fn? ifn?

Abstractions (Clojure type selection flowchart)

Protocols (clojure.org/reference/protocols)

Define	(defprotocol Slicey (slice [at]))
Extend	(extend-type String Slicey (slice [at] ...))
Extend null	(extend-type nil Slicey (slice [_] nil))
Reify	(reify Slicey (slice [at] ...))
Test	satisfies? extends?
Other	extend extend-protocol extenders

Records (clojure.org/reference/datatypes)

Define	(defrecord Pair [h t])
Access	(:h (Pair. 1 2)) → 1
Create	Pair. →Pair map→Pair
Test	record?

Types (clojure.org/reference/datatypes)

Define	(deftype Pair [h t])
Access	(.h (Pair. 1 2)) → 1
Create	Pair. →Pair
	(deftype Pair [h t]
With methods	Object
	(toString [this] (str "<" h " ", t ">")))

Multimethods (clojure.org/reference/multimethods)

Define	(defmulti my-mm dispatch-fn)
Method define	(defmethod my-mm :dispatch-value [args] ...)
Dispatch	get-method methods
Remove	remove-method remove-all-methods
Prefer	prefer-method prefers
Relation	derive underive isa? parents ancestors descendants make-hierarchy

Datafy (article)

Datafy	(clojure.datafy/) datafy nav
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Macros

Create	defmacro definline
Debug	macroexpand-1 macroexpand (clojure.walk/) macroexpand-all
Branch	and or when when-not when-let when-first if-not if-let cond condp case
	when-some if-some
Loop	for doseq dotimes while
Arrange	.. doto → →> as-> cond-> cond->> some-> some->>
Scope	binding locking time with-in-str with-local-vars with-open with-out-str
	with-precision with-redefs with-redefs-fn
Lazy	lazy-cat lazy-seq delay
Doc.	assert comment doc

Special Characters (clojure.org/reference/reader, guide)

,	Comma reads as white space. Often used between map key/value pairs for readability.
'	quote: 'form → (quote form)
/	Namespace separator (see Primitives/Other section)
\	Character literal (see Primitives/Other section)
:	Keyword (see Primitives/Other section)
;	Single line comment
~	Metadata (see Metadata section)
foo	'earmuffs' - convention to indicate dynamic vars, compiler warns if not dynamic
@	Deref: @form → (deref form)
'	Syntax-quote
foo#	'auto-gensym', consistently replaced with same auto-generated symbol everywhere inside same '(...)
-	Unquote
-@	Unquote-splicing
->	'thread first' macro →
->>	'thread last' macro →>
>!! <!! >! <!	core.async channel macros >!! <!! >! <!
(	List literal (see Collections/Lists section)
[	Vector literal (see Collections/Vectors section)
{	Map literal (see Collections/Maps section)
#'	Var-quote #'x → (var x)
##	##"p" reads as regex pattern p (see Strings/Regex section)
{:	Set literal (see Collections/Sets section)
#(	Anonymous function literal: #(...) → (fn [args] (...))
%	Anonymous function argument: %N is value of anonymous function arg N. % short for %1. %& for rest args.
##?	Reader conditional: ##?(:clj x :cljs y) reads as x on JVM, y in ClojureScript, nothing elsewhere. Other keys: :cljr :default
##?@	Splicing reader conditional: [1 ##?@(:clj [x y] :cljs [w z]) 3] reads as [1 x y 3] on JVM, [1 w z 3] in ClojureScript, [1 3] elsewhere.
#foo	tagged literal e.g. #inst #uuid
#:	map namespace syntax e.g. #:foo{:a 1 :b 2} is equal to {:foo/a 1 :foo/b 2}
##	symbolic values: ##Inf ##-Inf ##NaN
\$	JavaContainerClass\$InnerClass
foo?	conventional ending for a predicate, e.g.: zero? vector?
	instance? (unenforced)
foo!	conventional ending for an unsafe operation, e.g.: set! swap!
	alter-meta! (unenforced)
-	conventional name for an unused value (unenforced)
#_	Ignore next form

Metadata (clojure.org/reference/reader, special_forms)

General	~{:key1 val1 :key2 val2 ...}
Abbrevs	~Type → ~{:tag Type}, ~:key → ~{:key true}
Common	~:dynamic ~:private ~:doc ~:const
Examples	(defn ~:private ~String my-fn ...) (def ~:dynamic *dyn-var* val)
On Vars	meta with-meta vary-meta alter-meta! reset-meta! doc find-doc test

Special Forms (clojure.org/reference/special_forms)

def	if do let letfn quote var fn loop recur set! throw try monitor-enter
monitor-exit	
Binding Forms /	(examples) let fn defn defmacro loop for doseq if-let when-let
Destructuring	if-some when-some

Vars and global environment (clojure.org/reference/vars)

Def variants	def defn defn- definline defmacro defmethod defmulti defonce
	defrecord
Interned vars	declare intern binding find-var var
Var objects	with-local-vars var-get var-set alter-var-root var? bound?
	thread-bound?
Var validators	set-validator! get-validator

Namespace

Current	*ns*
Create/Switch	(tutorial) ns in-ns create-ns
Add	alias def import intern refer
Find	all-ns find-ns
Examine	ns-name ns-aliases ns-map ns-interns ns-publics ns-refers
	ns-imports
From symbol	resolve ns-resolve namespace the-ns (1.10) requiring-resolve
Remove	ns-unalias ns-unmap remove-ns

Loading

Load libs	(tutorial) require use import refer
List loaded	loaded-libs
Load misc	load load-file load-reader load-string

Concurrency

Atoms	atom swap! reset! compare-and-set! swap-vals! reset-vals!
Futures	future future-call future-done? future-cancel future-cancelled?
	future?
Threads	bound-fn bound-fn* get-thread-bindings push-thread-bindings
	pop-thread-bindings thread-bound?
Volatiles	volatile! vreset! vswap! volatile?
Misc	locking pcalls pvalues pmap seqe promise deliver

Refs and Transactions (clojure.org/reference/refs)

Create	ref
Examine	deref @ (@form → (deref form))
Transaction	sync dosync io!
In transaction	ensure ref-set alter commute
Validators	set-validator! get-validator
History	ref-history-count ref-min-history ref-max-history

Agents and Asynchronous Actions (clojure.org/reference/agents)

Create	agent
Examine	agent-error
Change state	send send-off restart-agent send-via set-agent-send-executor!
	set-agent-send-off-executor!
Block waiting	await await-for
Ref validators	set-validator! get-validator
Watchers	add-watch remove-watch
Thread handling	shutdown-agents
Error	error-handler set-error-handler! error-mode set-error-mode!
Misc	*agent* release-pending-sends

Java Interoperation (clojure.org/reference/java_interop)

General	.. doto Classname/ Classname. new bean comparator enumeration-seq
	import iterator-seq memfn set! class class? bases supers type
Cast	gen-class gen-interface definterface
	boolean byte short char int long float double bigdec bigint num cast
	biginteger
Exceptions	throw try catch finally pst ex-info ex-data Throwable->map
	StackTraceElement->vec (1.10) ex-cause ex-message (clojure.main/)
	ex-triage ex-str err->msg report-error
Streams	(1.12) stream-into! stream-reduce! stream-seq! stream-transduce!

Arrays

Create	make-array object-array boolean-array byte-array short-array char-array
	int-array long-array float-array double-array aclone to-array to-array-2d
	into-array
Use	aget aset aset-boolean aset-byte aset-short aset-char aset-int aset-long
	aset-float aset-double alength amap areduce
Cast	booleans bytes shorts chars ints longs floats doubles

Proxy (Clojure type selection flowchart)

Create	proxy get-proxy-class construct-proxy init-proxy
Misc	proxy-mappings proxy-super update-proxy

Zippers (clojure.zip/)

Create	zipper seq-zip vector-zip xml-zip
Get loc	up down left right leftmost rightmost
Get seq	lefts rights path children
'Change'	make-node replace edit insert-child insert-left insert-right
	append-child remove
Move	next prev
Misc	root node branch? end?

Other

XML	clojure.xml/parse xml-seq
REPL	*1 *2 *3 *e *print-dup* *print-length* *print-level* *print-meta*
	print-readably (1.12) (clojure.repl.deps/) add-lib add-lib sync-deps
Code	*compile-files* *compile-path* *file* *warn-on-reflection* compile
	loaded-libs test
Misc	eval force hash name *clojure-version* clojure-version
	command-line-args (1.11) random-uuid
Browser / Shell	(clojure.java.browse/) browse-url (1.12) (clojure.java.process/) exec exit-ref
	from-file start stderr stdin stdout to-file